

Project Proposal

Live Classroom and Quiz Platform

Field	Details
Course	Database Systems
Submitted To	Sir Talha Shahid
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Group Members

Name	Student ID	Role
Muhammad Maaz	24K-0968	Student Web Application
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1. Introduction

Modern education increasingly demands real-time interaction and instant feedback mechanisms between teachers and students. Traditional classroom assessments are static, slow to grade, and fail to engage the digital-native student. This project proposes the development of a **Live Classroom and Quiz Platform** — a full-stack, web-based application that enables teachers to create courses, design quizzes, and launch interactive live quiz sessions, while students can enroll in courses, receive real-time notifications, and participate in live quizzes with instant scoring and leaderboards.

The platform is split into two independent but interconnected web applications sharing a common backend:

- **Teacher Web App** — for course and quiz management, and live session control.
 - **Student Web App** — for course enrollment, quiz participation, and performance tracking.
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2. Problem Statement

Current educational tools suffer from several critical limitations:

- **Lack of real-time interaction:** Traditional quiz tools (paper-based or asynchronous online tools) do not support live, synchronized quiz sessions.
- **No immediate feedback:** Students must wait to receive grades and results, reducing motivation.
- **Disjointed systems:** Teachers manage attendance, quizzes, and grading in separate tools with no unified platform.
- **No instant notification:** Students are not instantly alerted when a live session begins, causing them to miss sessions.

- **Poor engagement:** Without live leaderboards and competitive elements, student engagement in online quizzes is low.
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3. Proposed Solution

We propose building **ClassHub** — a dual-application web platform that solves each of the above problems through modern real-time web technologies.

3.1 Teacher Web Application

The teacher-facing application will provide:

- **Authentication:** Secure sign-in / sign-up with role-based access control (teachers only).
- **Course Management:** Create, view, and manage multiple courses with title and description.
- **Quiz Builder:** Add quizzes to courses with configurable questions (Multiple Choice, True/False, Short Answer), point values, and time limits per question.
- **Session Control:** Launch live quiz sessions, advance questions in real-time, and end sessions.
- **Notification System:** Automatically notify enrolled students via **Push Notifications (Firebase Cloud Messaging)** and **Email (Gmail / Nodemailer)** when a live quiz starts.

3.2 Student Web Application

The student-facing application will provide:

- **Authentication:** Secure sign-in / sign-up with role-based access control (students only).
 - **Profile Setup:** Students set up their display name and profile avatar.
 - **Course Enrollment:** Enroll in a course using a course ID provided by the teacher.
 - **Live Quiz Participation:** Join a live session, view real-time questions, answer within a countdown timer, and auto-submit if the timer expires.
 - **Live Leaderboard:** See a real-time leaderboard after each question, showing all students' scores.
 - **Performance Tracking:** View personal quiz history and progress over time.
 - **Push Notifications:** Browser push notifications alerting students when a quiz goes live.
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4. Technical Architecture

4.1 Technology Stack

Layer	Technology	Justification
Frontend	Next.js 16 (App Router), React 19	Server-side rendering, file-based routing, TypeScript
Styling	Tailwind CSS v4 + Vanilla CSS	Rapid utility styling with custom glassmorphism UI

Layer	Technology	Justification
Backend / API	Next.js API Routes (Route Handlers)	Server-side logic co-located with the UI
Database	Supabase (PostgreSQL)	Managed DB with Row-Level Security (RLS)
Authentication	Supabase Auth	Email/password auth with SSR session handling
Real-time	Supabase Realtime (Presence + Postgres)	WebSocket-based live updates without a separate server
Push Notifs	Firebase Cloud Messaging (FCM)	Cross-platform browser push notifications
Email	Nodemailer + Gmail SMTP	Sending quiz alert emails to enrolled students
Deployment	Vercel	Zero-config deployment with serverless functions

4.2 System Architecture Diagram



4.3 Database Schema (Key Tables)

Table	Key Columns
users	id, role (teacher/student), full_name, avatar_url, fcm_token
courses	id, teacher_id, title, description, created_at
enrollments	id, course_id, student_id
quizzes	id, course_id, title, time_limit_seconds
questions	id, quiz_id, question_text, type, options, correct_answer, points, order_index
sessions	id, quiz_id, teacher_id, status, current_question_index, started_at
answers	id, session_id, question_id, student_id, answer_text, is_correct

5. Key Features Summary

Feature	Teacher App	Student App
Email/Password Authentication	YES	YES
Role-based Access Control	YES	YES
Course Creation & Management	YES	NO
Quiz Builder (MCQ/T-F/SA)	YES	NO
Course Enrollment	NO	YES
Profile & Avatar Setup	NO	YES
Launch Live Sessions	YES	NO
Real-time Question Control	YES	NO
Live Quiz Participation	NO	YES
Countdown Timer per Question	NO	YES
Auto-submit on Timeout	NO	YES
Real-time Leaderboard	YES	YES
Push Notifications (FCM)	YES (send)	YES (receive)
Email Notifications	YES (send)	YES (receive)
Lobby with Live Presence	NO	YES
Score & Progress Tracking	NO	YES

6. Project Division of Work

Component	Responsible Member
Teacher Web App (all pages & logic)	Sharjeel Ahmed (24K-0724)
Student Web App (all pages & logic)	Muhammad Maaz (24K-0968)
Supabase Database Schema & RLS Policies	Both Members
Firebase FCM Integration	Both Members
Email Notification (Nodemailer)	Sharjeel Ahmed (24K-0724)
Deployment (Vercel)	Both Members

7. Deployment Plan

Both applications will be deployed as separate projects on **Vercel**, each with their own domain:

- **Teacher App:** <https://live-classroom-and-quiz-platform-teacher-web.vercel.app/>
- **Student App:** <https://live-classroom-and-quiz-platform-student-web.vercel.app/>

Both apps share the same **Supabase** project and **Firebase** project as the common backend.

Supabase Authentication is configured with both app domains as allowed redirect URLs to support multi-domain SSR session handling.

8. Timeline

Week	Milestone
1	Project setup, Supabase schema design, authentication setup
2	Teacher App: Course management + Quiz builder
3	Student App: Enrollment, dashboard, profile setup
4	Real-time quiz sessions (teacher control + student participation)
5	Notifications (FCM push + email), leaderboard, progress tracking
6	Testing, bug fixes, UI polish, deployment

9. Expected Outcomes

By the end of this project, we will have delivered:

1. A fully deployed, production-ready dual web application.
 2. A complete real-time quiz experience from session creation to final leaderboard.
 3. An automated multi-channel notification system (push + email) alerting students of live quizzes.
 4. A polished, premium dark glassmorphism UI with smooth animations.
 5. Secure, role-separated access for teachers and students with Supabase RLS.
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10. Conclusion

The **Live Classroom and Quiz Platform** addresses a genuine need in modern education by combining real-time communication, gamified assessment, and multi-channel notifications into a single, cohesive platform. The proposed solution leverages industry-standard technologies and a well-defined architecture to ensure scalability, security, and a premium user experience. This project will demonstrate our ability to build and deploy a full-stack, production-grade web application as a team.